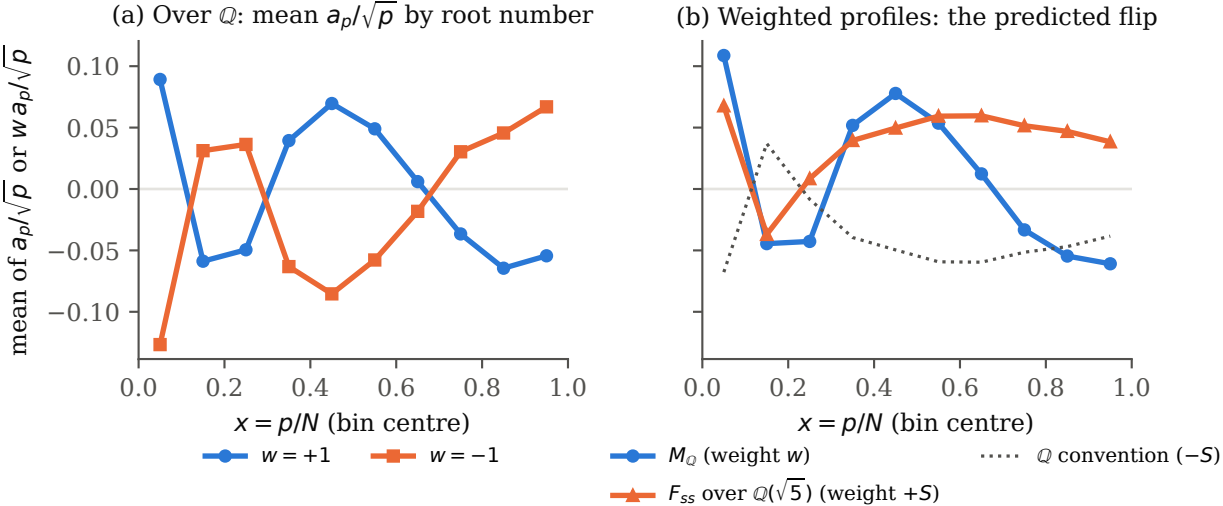
(a) Over $\mathbb{Q}$: mean $a_p/\sqrt{p}$ by root number
mean of $a_p/\sqrt{p}$ or $w\,a_p/\sqrt{p}$
0.10
0.05
0.00
−0.05
−0.10
0.0
0.2
0.4
0.6
0.8
1.0
$x = p/N$ (bin centre)
$w = +1$
$w = -1$
(b) Weighted profiles: the predicted flip
$x = p/N$ (bin centre)
$M_{\mathbb{Q}}$ (weight w)
$\mathbb{Q}$ convention ($-S$)
F_{ss} over $\mathbb{Q}(\sqrt{5})$ (weight $+S$)